

SNP × BMI Interaction Effects on Knee Osteoarthritis Risk: A Gene-Environment Interaction Analysis in UK Biobank

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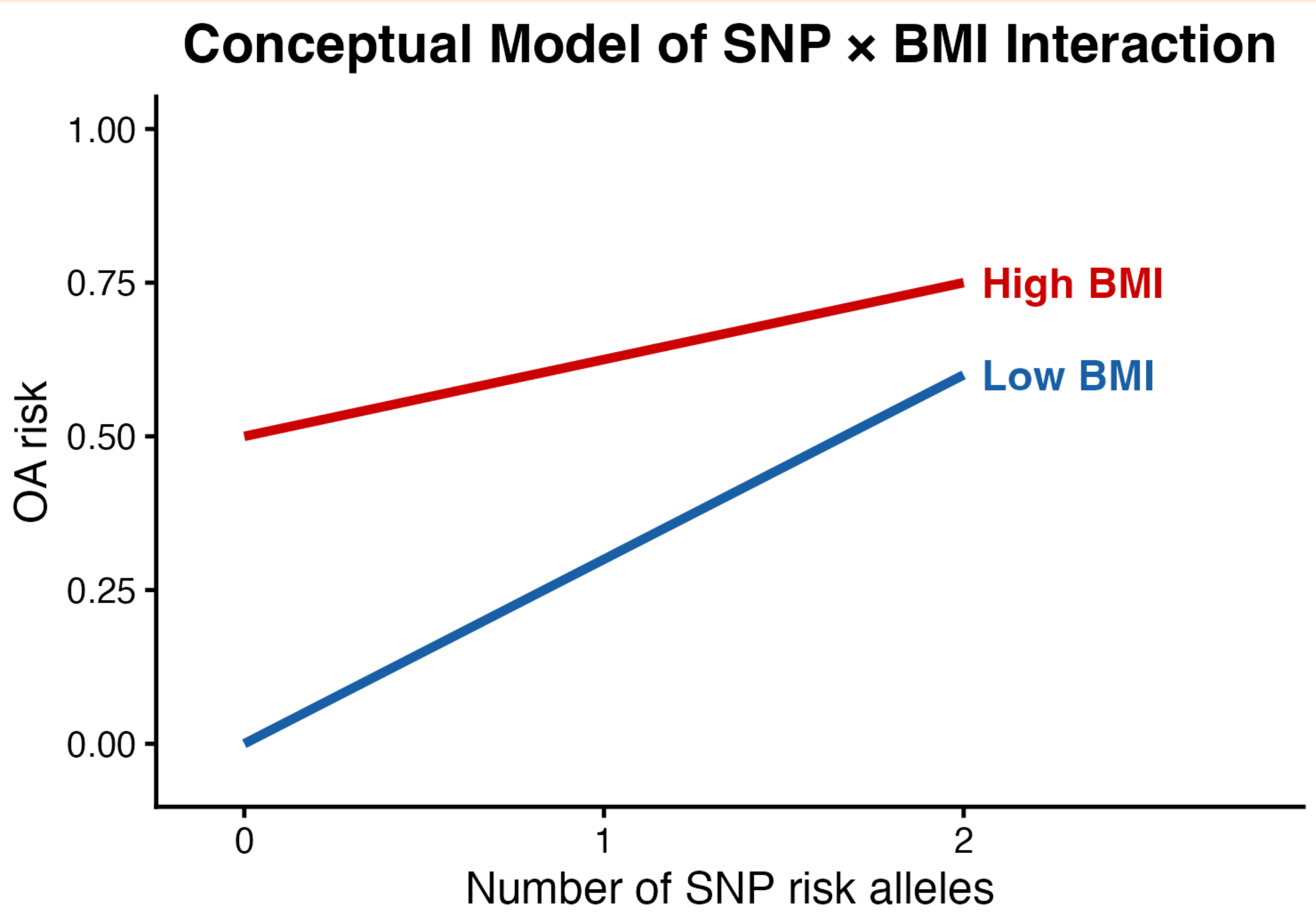


Background:

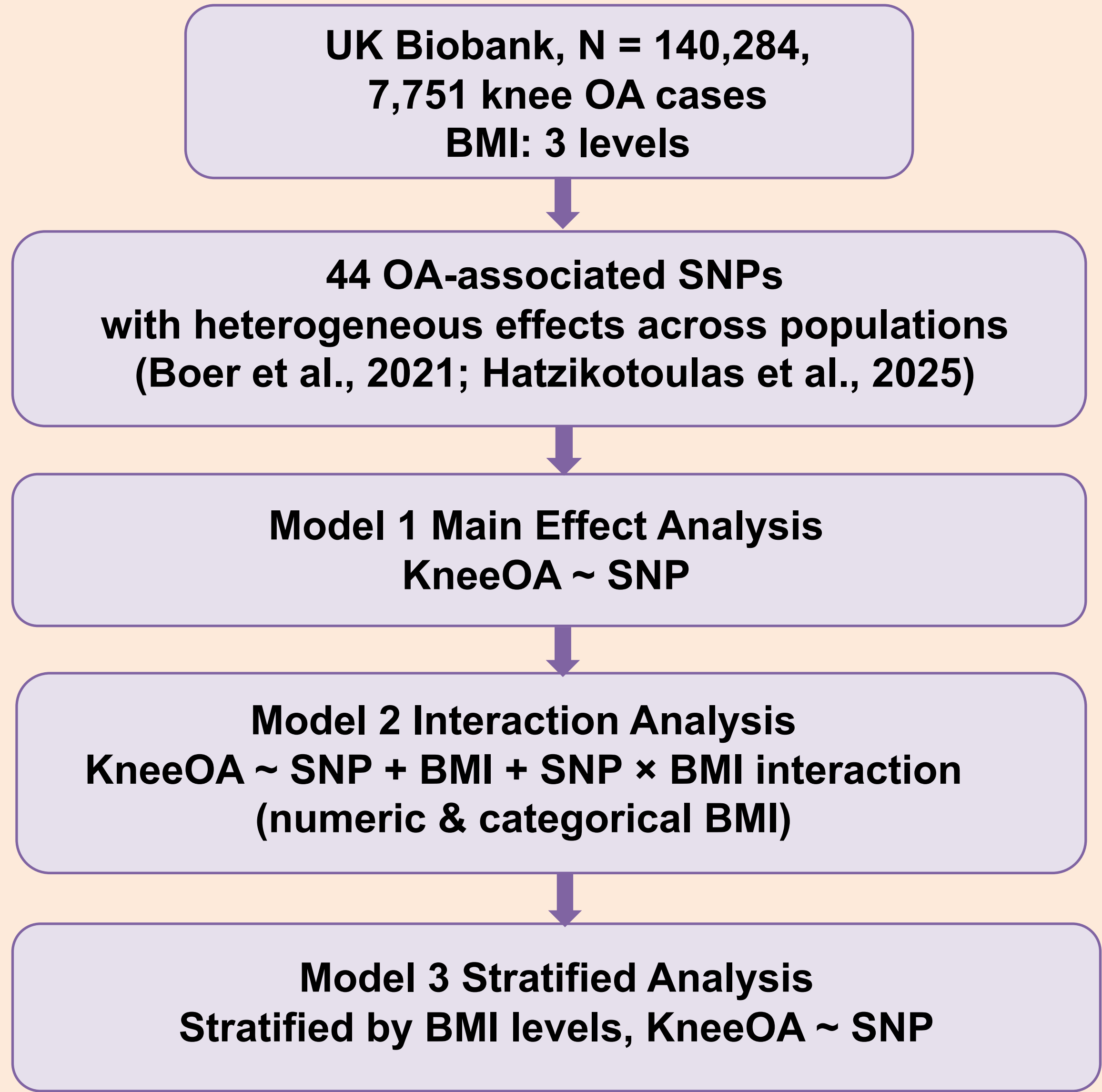
- Knee OA is a leading cause of disability
- BMI is one of the strongest modifiable risk factors
- The heterogeneous effects of OA-associated SNPs across populations suggests that gene-environment interactions may influence risk, which motivates our study.

Research question :

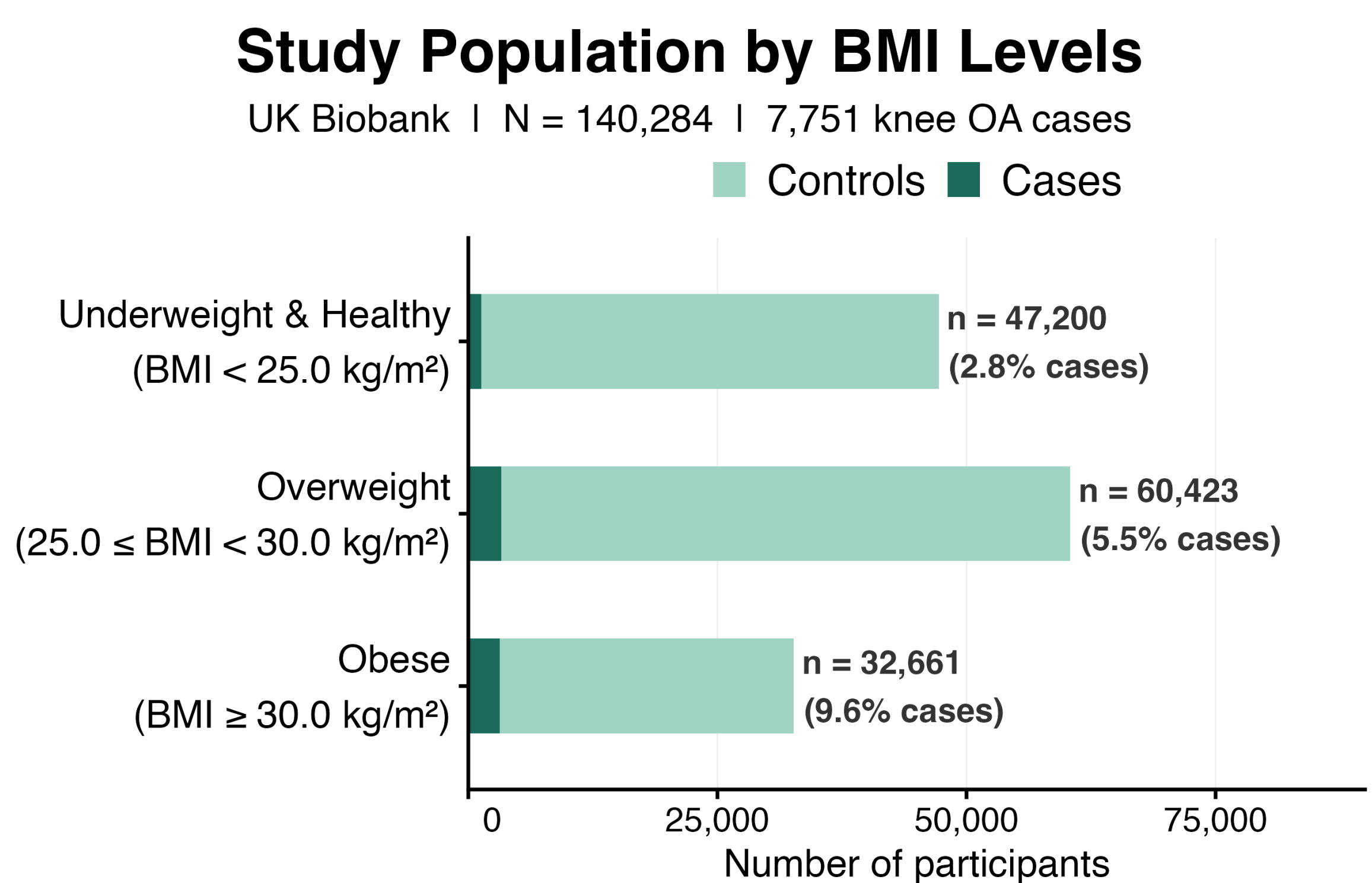
Do knee OA risk variants exert different effects across BMI levels?



Method:



All analyses are adjusted for age, sex, PC1-10;
Bonferroni correction: $P < 0.00114$ (0.05 / 44 SNPs)



SNP × BMI Interaction and Main Effect Results

Among 44 candidate SNPs, three variants near *COL27A1* and one variant near *BTNL2* showed nominal SNP × BMI interaction ($P < 0.05$). Two SNPs were selected as stable references: rs4144005 in *ALDH1A2* (a known OA risk gene) and rs1421085 in *FTO* (a canonical BMI-associated gene). Both showed significant main effects but no SNP × BMI interaction.

Main effects and SNP×BMI interaction results ¹								
6 selected SNPs from targeted analysis of 44 knee OA-associated loci; UK Biobank, N = 140,284								
SNP	Reported Gene	Chromosome	Model 2: SNP×BMI Interaction			Model 1: Main Effect		Functional Annotation ⁴
			Continuous SNP×BMI P ²	Categorical SNP×BMI P (OW vs UH) ²	Categorical SNP×BMI P (OBE vs UH) ²	OR (95% CI) ³	Main effect P ³	
Interaction signal: SNPs with nominal SNP×BMI interaction (P<0.05)								
rs2043195	COL27A1, KIF12	9	0.0025	0.5252	0.0071	1.039 (1.005–1.075)	0.0260	Regulatory region variant near COL27A1 (known OA risk gene)
rs1078301	COL27A1	9	0.0014	0.3519	0.0036	1.039 (1.001–1.078)	0.0440	Intergenic variant near COL27A1 (known OA risk gene); most significant variant for this locus (PIP=0.46)
rs919642	COL27A1	9	0.0023	0.3510	0.0052	1.038 (0.999–1.077)	0.0539	Intergenic variant near COL27A1 (known OA risk gene)
rs10947262	BTNL2	6	0.0082	0.8602	0.0296	0.964 (0.904–1.028)	0.2605	Intron variant in BTNL2 in the HLA class II region; associated with knee OA in a Japanese GWAS (Nakajima et al., 2010); failed to replicate in large European-descent cohorts (Valdes et al., 2011)
Stable reference: SNPs without significant SNP×BMI interaction; main effects on knee OA risk remain significant								
rs4144005	ALDH1A2	15	0.9305	0.1233	0.6521	0.931 (0.901–0.962)	2.20e-05	Intron variant in ALDH1A2 (known OA risk gene)
rs1421085	FTO	16	0.3352	0.7664	0.3516	1.060 (1.026–1.096)	5.50e-04	Intron variant in FTO (canonical BMI-associated gene)

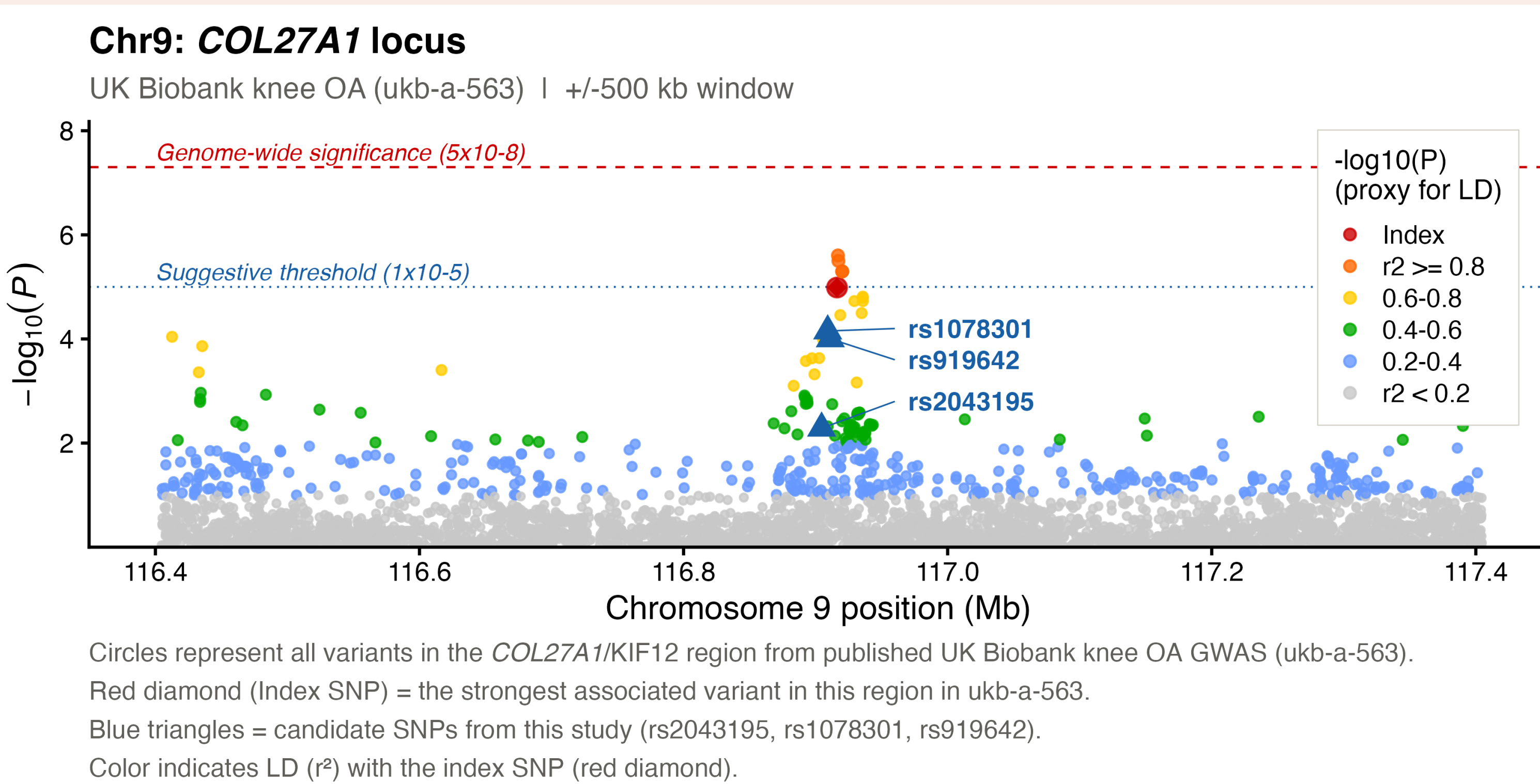
¹ Bonferroni-corrected threshold: P < 0.00114 (0.05/44 SNPs). No SNP reached Bonferroni significance for the interaction test. Red bold = P < 0.05.

² UH = underweight & healthy (BMI < 25.0 kg/m²); OW = overweight (25.0 ≤ BMI < 30.0 kg/m²); OBE = obese (BMI ≥30.0 kg/m²). Two interaction terms shown separately: BMI category was modeled as a nominal variable (dummy-coded, reference = underweight & healthy).

³ Adjusted for age, sex, and PC1–10, without BMI adjustment.

⁴ Functional annotation from Hatzikotoulas et al., Nature 2025, unless otherwise noted.

Genomic context: COL27A1 as a known OA risk locus

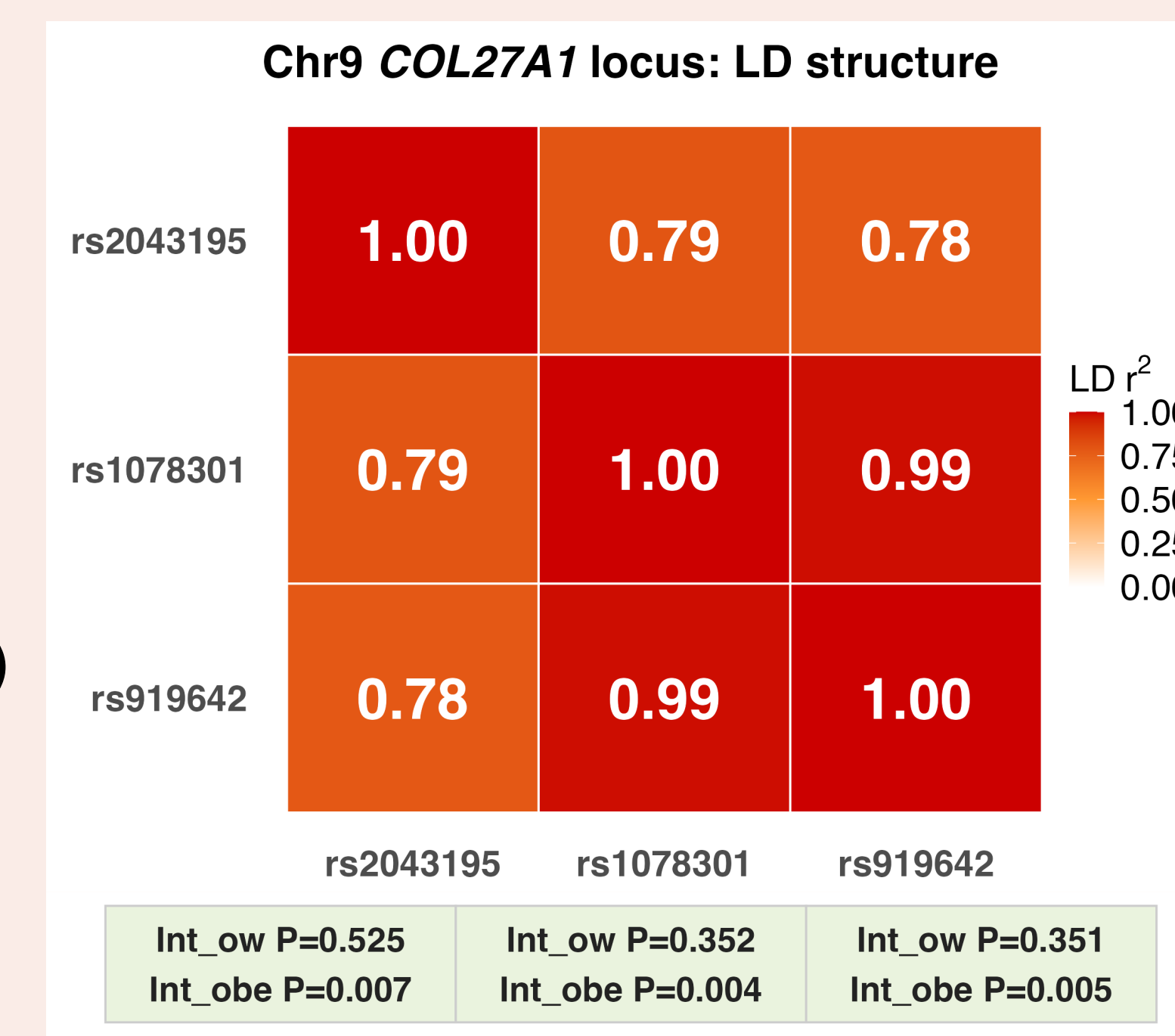
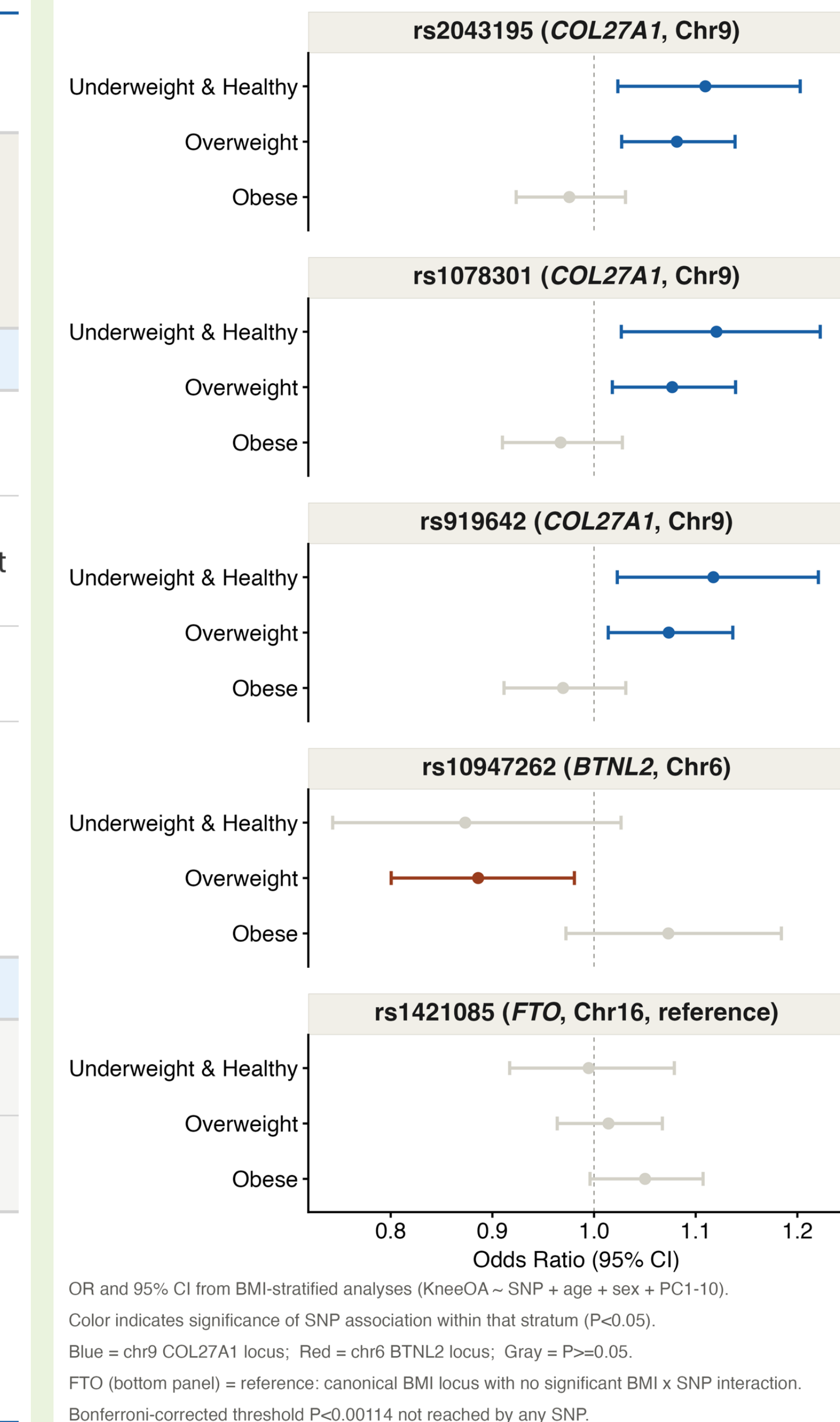


The *COL27A1* locus at chr9:116.9 Mb shows established association with knee OA in the published UK Biobank GWAS (ukb-a-563; Locus Zoom), supporting the biological relevance of our interaction signal.

The three candidate SNPs (rs2043195, rs1078301, rs919642) form a single tight LD block (r²=0.78-0.99; LD Heatmap), indicating they represent one shared genetic signal.

BMI-stratified odds ratios (Forest Plot)

COL27A1 (chr9) risk effects were significant in the underweight & healthy and overweight strata but attenuated to null in the obese group; *BTNL2* (chr6) showed a protective effect only in the overweight stratum.



Conclusions:

No SNP × BMI interaction reached Bonferroni threshold ($P < 0.00114$); results are suggestive and warrant further investigation.

Other observations of interest (suggestive and not to be over-interpreted at this stage):

- For SNPs in both *COL27A1* and *BTNL2*, the risk alleles are only associated with elevated knee OA risk in non-obese individuals:** In obese individuals, OA susceptibility may be predominantly driven by non-genetic factors (e.g., mechanical load), which may obscure the detectable genetic signal.
The *COL27A1* interaction with BMI is of interest because it is a high-confidence OA effector gene in the literature. (chondrocyte Hi-C, eQTL PIP=0.46, human & mouse MSK phenotype; Hatzikotoulas et al., 2025)
- This pattern is not universal:** *FTO* (a canonical BMI-associated gene) and *ALDH1A2* (a well-established OA risk gene) both showed stable effects across all BMI strata.

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References:

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